SPECIFICATION FOR
"UD" SINGLE POLE DISTRIBUTION BLOCKS
or engineering approved equivalent per the specification below

1. SUMMARY

This specification covers the technical requirements of the UD single pole distribution blocks for use in low-voltage power distribution applications where electrical connections between live parts are required.

2. COMPLIANCE REQUIREMENTS

a. ANSI/UL1059 "Terminal Blocks" (Recognized by Underwriters Laboratories under this category) or,
b. ANSI/UL1953 "Terminal Blocks" (Listed by Underwriters Laboratories under this category)
c. CSA C22.2 NO. 158-10 "Terminal Blocks"
d. IEC 60947-7-1
e. Ui = 1000V AC/DC IEC minimum 600 V UL or 1000V UL in function of the model
f. Short circuit current rated up to 100 KA – As per UL
g. RoHS 2002/95/EC Compliant
h. CE marked
i. EAC certificate or Customs Union certificate
j. Flame retardant as per UL 94 V-0
k. Flame retardant as per IEC® 60695-2-11 (Glow Wire Test 960 °C)
l. Low Smoke
m. Halogen Free

3. PRODUCT COMPOSITION

a. Connection block and connection terminals (screws)

The connection block should be made of tinned copper or tinned Aluminum and the connection terminals (screws) be made of low carbon steel with tin plating.

The design of the connection block/connection terminal should yield a 95% fill ratio* and allow for connection to a solid or standard cable with or without a ferule or to a busbar, to a Flexible Busbar or Braid with Solid Palm.

The design should allow for visual inspection of the wire and confirmation of connection. It should also allow for copper and aluminum cables to be connected on the same block without any risk of galvanic corrosion.

*the fill ratio is the ratio between the major diameter of the connection terminal and the size of the hole in the connection block itself.

b. Housing

The housing should be made of self-extinguishable resin rated to class V0 according to UL 94. It should be halogen free.

The housing should be design to be connected onto a DIN rail and should have holes on the backside for a direct connection to a panel. It should also be stackable with other similar distribution blocks. DIN rail clip should be a 2 positions design (open/close)

c. Cover

The cover should be made of self-extinguishable material rated to class V0 according to UL 94, be transparent and hinged to the housing or removable therefore providing direct access to the connection terminals (screws) for maintenance purposes. It should include a screw retaining feature, be halogen free and marked with the operating current and operating voltage of the distribution block according to IEC and UL.

d. Jumper

An optional jumper and plug should be provided as an option to the 125A (IEC rating) and 160A (IEC rating). This jumper should be made of tinned copper with self-extinguishable insulating sleeve rated to class V0 according to UL 94 on it. It should be halogen free.

4. PRODUCT CHARACTERISTICS

a. Physical

The assembly should be IP20 finger safe.

b. Environmental

Ambient temperature ranges

Applications per IEC 60947-7-1: -5°C to 40°C (with allowable temperature rise of 45K)

Applications per UL 1059: -40°C to 40°C (with allowable temperature rise of 30K)

Applications per CSA C22.2 No. 158-10: -40°C to 60°C (with allowable temperature rise of 30K)

Derating according to Ambient* Temperature (°C) to maintain working temperature of 85°C									
Ambient Temperature (°C)	30°	35°	40°	45°	50°	55°	60°	65°	70°
Derating Coefficient (d)	1	1	1	0.94	0.88	0.82	0.75	0.67	0.58
*environment around the terminal blocks inside the enclosure									

c. Performance

The blocks should be tested per IEC 60974-7-1 and to UL1059, or UL1953

The maximum operating voltage should be 1000V AC/DC per IEC and 600V or 1000V per UL.

The performance for each block of a certain rating should be as follow:

Max IEC Current	Max UL Current	Line side : Nbre of connection	Line side Min and Max conductor size	Load side : Nbre of	Load side Min and Max conductor size	Max working voltage IEC	Max working voltage UL	IEC Icw for 1sec (kA RMS)	IEC IPk (kA)	UL SCCR (kA RMS)
80 A	85 A	1 Cable	6 - 16 mm² #16 - # 4 AWG	6 Cables	2.5 - 16 mm² #16 - # 4 AWG	1,000 VAC/DC	600 VAC/DC	3	22	100
125 A	150 A	1 Cable	10 - 35 mm² #8 - 1/0 AWG	7 Cables	2.5 - 16 mm² #14 - # 4 AWG	1,000 VAC/DC	600 VAC/DC	4,2	30	100
160 A	200 A	1 Cable	10 - 70 mm² #8 - 3/0 AWG	7 Cables	2.5 - 16 mm² #14 - # 4 AWG	1,000 VAC/DC	600 VAC/DC	11,8	30	100
250 A	255 A	1 Cable	35 - 120 mm² #6 AWG - 250 kcmil	11 Cables	2.5 - 35 mm² #14 - # 1 AWG	1,000 VAC/DC	600 VAC/DC	24,5	51	100
250 A	255 A	Flat Conductor	Flexibar 3X9X0.8 - 6x15.5x0.8	6 Cables	2.5 - 16 mm² #14 - # 4 AWG	1,000 VAC/DC	600 VAC/DC	23	23	100
400 A	335 A	1 Cable	95 - 185 mm² 3/0 AWG - 400 kcmil	11 Cables	2.5 - 35 mm² #14 - # 1 AWG	1,000 VAC/DC	600 VAC/DC	24,5	51	100
400 A	335 A	1 Cable	95 - 185 mm² 3/0 AWG - 400 kcmil	12 Cables	2.5 - 10 mm² # 14 - # 6 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	24,5	51	100
400 A	335 A	1 Cable	96 - 185 mm² 3/0 AWG - 400 kcmil	12 Cables	2.5 - 10 mm² # 14 - # 6 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	24,5	51	100
400 A	400 A	2 Cables	35 - 95 mm² #8 - 3/0 AWG	12 Cables	2.5 - 10 mm² # 14 - # 6 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	24,5	51	100
400 A	400 A	2 Cables	36 - 95 mm² #8 - 3/0 AWG	12 Cables	2.5 - 10 mm² # 14 - # 6 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	24,5	51	100
500 A	335 A	Flat Conductor	Flexibar 4x15.5x0.8 - 8x24x1	11 Cables	2.5 - 35 mm² #14 - # 1 AWG	1,000 VAC/DC	600 VAC/DC	24,5	51	10
500 A	380 A	1 Cable	95 - 240 mm² 3/0AWG- 500 kcmil	6 Cables	10 - 50 mm² #8 - 1/0 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	34,3	52,5	100
500 A	475 A	Flat Conductor	Flexibar 2x20x1 - 10x24x1 IBS/IBSB 50 - 100 mm²	6 Cables	10 - 50 mm² #8 - 1/0 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	34,3	52,5	100
500 A	490 A	Flat Conductor	Flexibar 2x20x1 - 10x24x1 IBS/IBSB 50 - 100 mm²	9 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	32,2	52,5	100
500 A	500 A	Flat Conductor	Flexibar 2x20x1 - 10x24x1 IBS/IBSB 50 - 100 mm²	12 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	34,3	52,5	100
630 A	420 A	1 Cable	120 - 300 mm² 4/0 AWG- 600 kcmil	9 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	32,2	52,5	100
630 A	670 A	2 Cables	95 - 185 mm² 3/0 AWG - 400 kcmil	12 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	42,9	52,5	100
800 A	670 A	Flat Conductor	Flexibar 2x20x1 - 10x32x1 IBS/IBSB 50 - 240 mm²	12 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	42,9	73,5	100
1000 A	760 A	2 Cables	35 - 240 mm² 2 AWG - 500 kcmil	12 Cables	4 - 25 mm² #12 - # 4 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	42,9	73,5	100
1000 A	840 A	Flat Conductor	Flexibar 6x24x1 - 10x50x1 IBS/IBSB 120 - 240 mm²	9 Cables	10 - 95 mm² #8 - 3/0 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	71,5	73,5	100
1250 A	950 A	2 Cables	185 - 400 mm² 400 kcmil - 750 kcmil	9 Cables	10 - 95 mm² #8 - 3/0 AWG	1,000 VAC, 1,500 VDC	1,000 VAC/DC	84	73,5	100

* Connection of the rectangular shape input conductor to the connection block should not require the input conductor to be drilled or punch. Connection of the rectangular shape conductor need to keep the IP20 rating with any size of conductor.

5. SUPPLIER'S QUALIFICATION AND QUALITY CONTROL

a. Supplier shall be ISO9001:2000 certified and quality control be done accordingly.
b. Supplier shall be following a health & safety program at least as stringent as the United States Occupational Health & Safety Administration program.